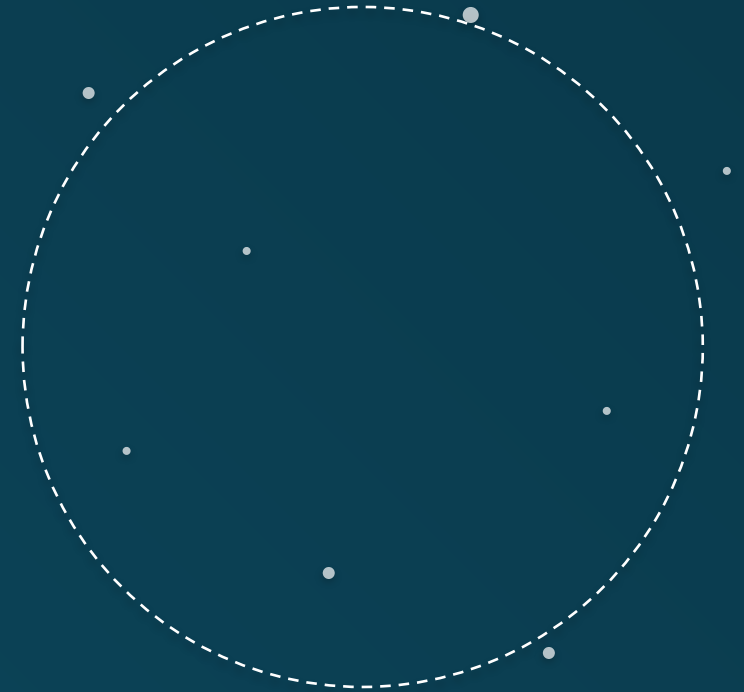


TOPIC 25

Astronomy & Cosmology

天文与宇宙学



CIE 9702 · A Level Physics A2 · Paper 4

剑桥国际物理 A2 分章节讲义 · 中英双语

Learning Outcomes

学习目标

LEARNING OUTCOMES

学习目标

By the end of this topic, you will
be able to...

25.1 Luminosity and radiant flux 光度与辐射通量

- Luminosity L and radiant flux intensity $F = L/(4\pi d^2)$ 光度与通量
- Standard candles: Cepheid variables 标准烛光与造父变星

25.2 Stellar radii 恒星半径

- Wien's displacement law $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$ 维恩位移定律
- Stefan-Boltzmann law $L = 4\pi\sigma r^2 T^4$ 斯忒藩 - 玻尔兹曼定律
- Combine the two laws to estimate stellar radii 联立估算半径

25.3 Hubble's law and the Big Bang 哈勃定律与大爆炸

- Redshift: $\Delta\lambda/\lambda \approx v/c$ 红移
- Hubble's law $v \approx H_0 d$; expanding Universe 哈勃定律
- Age of the Universe $t \approx 1/H_0$ 宇宙年龄

Luminosity & Radiant Flux

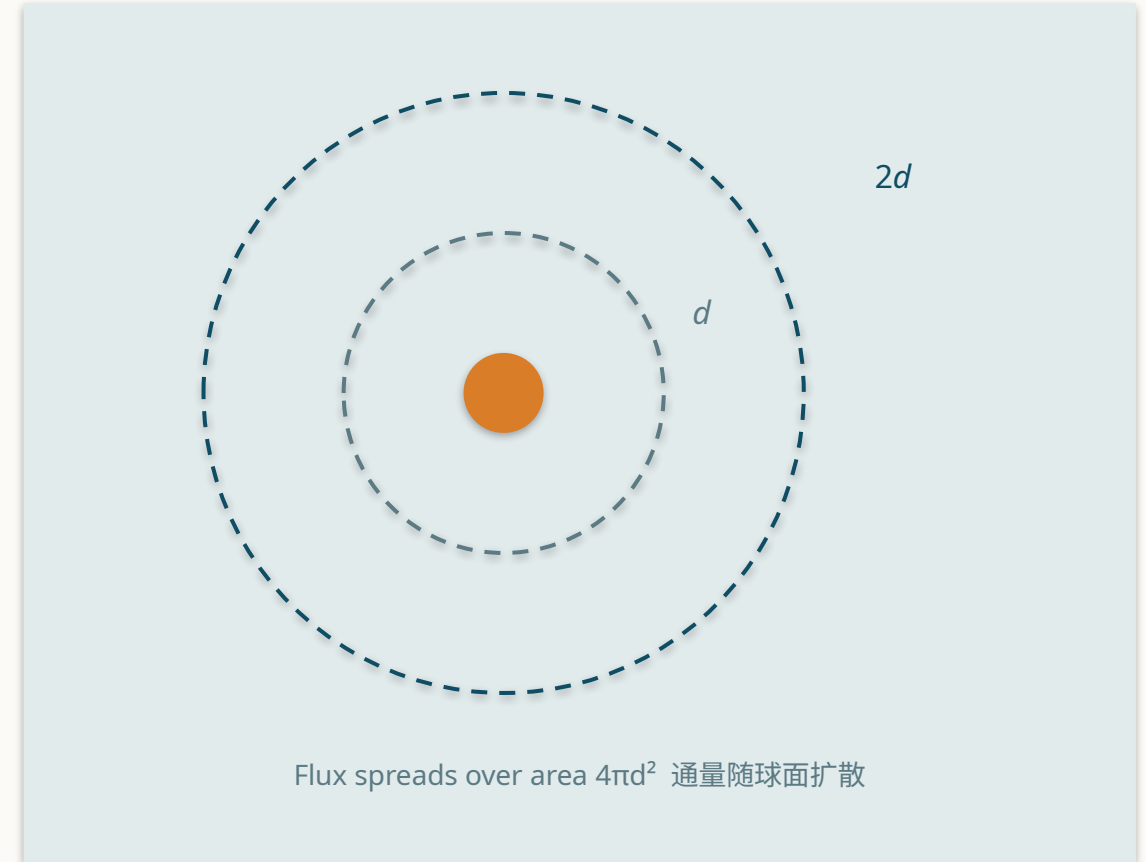
光度与辐射通量

The **luminosity** L of a star is its total radiated power (W).
The **radiant flux intensity** F is the radiant power per unit area received at distance d (W m^{-2}).

光度 = 恒星辐射总功率；辐射通量 = 单位面积接收功率。

$$F = \frac{L}{4\pi d^2} \quad \text{反平方定律}$$

- Radiation spreads over a sphere of area $4\pi d^2$ 球面扩散
- F is what we measure as "apparent brightness" 表观亮度即 F
- Same star at double distance $\Rightarrow$ quarter of the flux 距离加倍通量减为 1/4



Standard Candles

标准烛光与造父变星

A **standard candle** is a class of stellar object of known luminosity, whose distance can be calculated from its measured radiant flux intensity and its luminosity.

光度已知的一类天体；由 L 与测得的 F 求距离。

- **Cepheid variables:** luminosity varies periodically (1–100 days) 造父变星周期性变光
- Period–luminosity relation: measure the period $\Rightarrow$ know L 周期 - 光度关系
- Method: measure period $\Rightarrow L$; measure F ; compute 测距三步
- Reaches distances of millions of light-years; $1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$ 适用遥远星系

Wien's Displacement Law

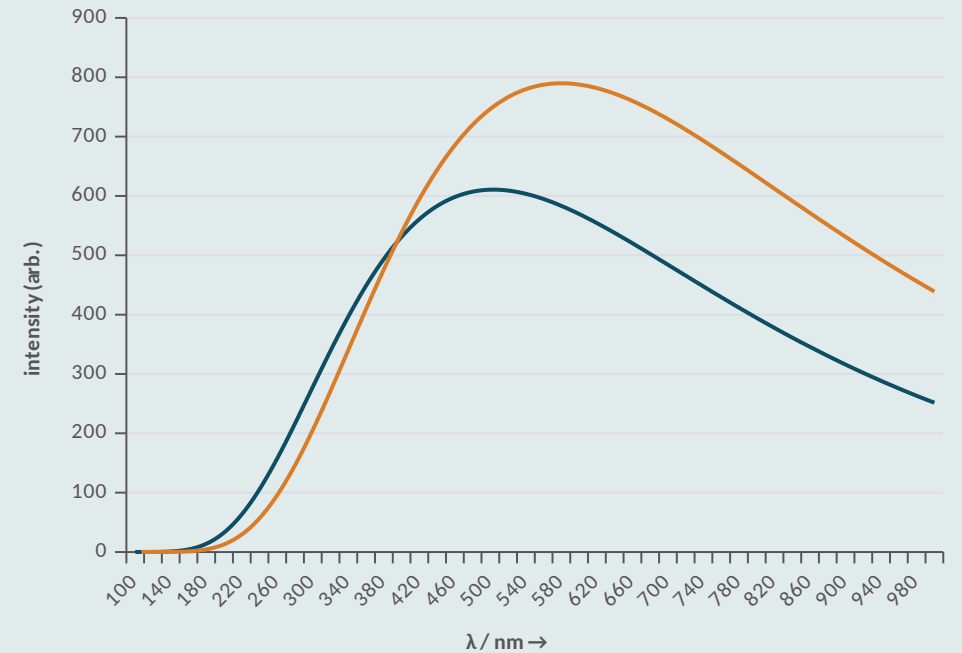
维恩位移定律

Stars behave approximately as **black bodies**: their emission spectrum peaks at a wavelength λ_{max} which depends only on surface temperature T .

恒星近似黑体辐射，谱峰波长只与表面温度有关。

$$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K} \quad T \text{ 必须用开尔文}$$

- Hotter star $\Rightarrow$ shorter λ_{max} (bluer); cooler $\Rightarrow$ longer (redder) 越热越蓝
- Measure $\lambda_{\text{max}} \Rightarrow$ surface temperature 测峰值得温度
- The Sun: $\lambda_{\text{max}} \approx 510 \text{ nm} \Rightarrow T \approx 5700 \text{ K}$ 太阳示例



Black-body spectra at two temperatures 两个温度的黑体谱

Stefan-Boltzmann Law

斯忒藩 - 玻尔兹曼定律

The luminosity of a spherical black body of radius r at temperature T is proportional to its surface area and to the fourth power of T .

光度正比于表面积与温度的四次方。

$$L = 4\pi\sigma r^2 T^4, \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

斯忒藩 - 玻尔兹曼常量

- Luminosity is set by **both size and temperature** L 由 r 与 T 共同决定
- T^4 makes L extremely sensitive to temperature T^4 敏感
- A red giant can outshine a small hot star by its huge r^2 红巨星靠体积取胜
- Unit check: $\text{m}^2 \times \text{K}^4 \times \text{W m}^{-2} \text{ K}^{-4} = \text{W}$ 量纲自检

Estimating Stellar Radii

估算恒星半径

1. **Measure λ_{max}** from the spectrum $\Rightarrow$ Wien's law gives surface temperature T 谱峰 $\Rightarrow$ 温度
 2. **Find L** from a standard-candle method (e.g. Cepheid period + measured F) 标准烛光 $\Rightarrow$ 光度
 3. **Solve $L = 4\pi\sigma r^2 T^4$** for r 联立求半径
- - Only an estimate — luminosity itself is estimated 结果是估算值
 - Compare with the Sun: $r_{\odot} = 7.0 \times 10^8 \text{ m}$, $L_{\odot} = 3.9 \times 10^{26} \text{ W}$ 太阳参考值
 - Giant stars: r up to hundreds of $r_{\odot}$; white dwarfs $\approx$ Earth-sized 量级感受

Redshift

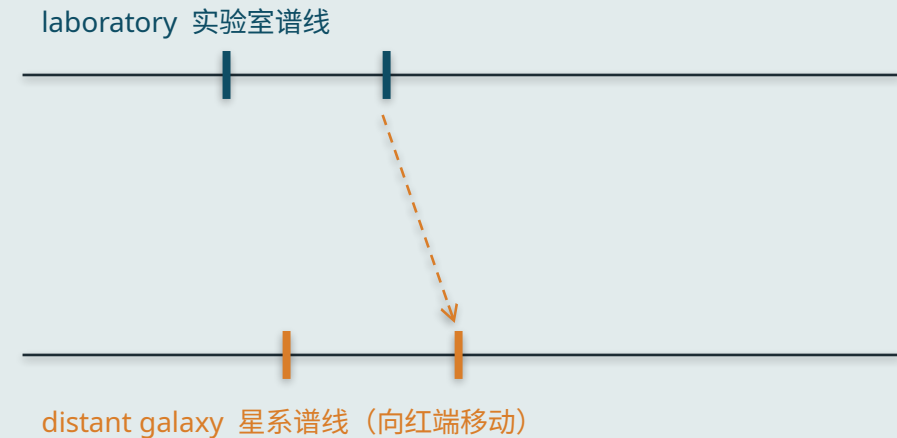
红移

Spectral lines from distant galaxies are shifted to **longer wavelengths** (redshift). This is a Doppler-type effect: the galaxy is receding from us.

遥远星系谱线波长变长 = 红移，表明星系正在退行。

$$z = \frac{\Delta \lambda}{\lambda} \approx \frac{v}{c} \quad (v \ll c) \quad \text{低速近似； } z \text{ 为红移量}$$

- $\Delta \lambda > 0$: redshift, object receding; $\Delta \lambda < 0$: blueshift, approaching
红移退行、蓝移接近
- Compare galaxy lines with laboratory wavelengths 与实验室波长对比
- Almost all galaxies show redshift $\Rightarrow$ the Universe is expanding
宇宙膨胀证据



Lines shifted to longer λ 谱线红移对比

Hubble's Law

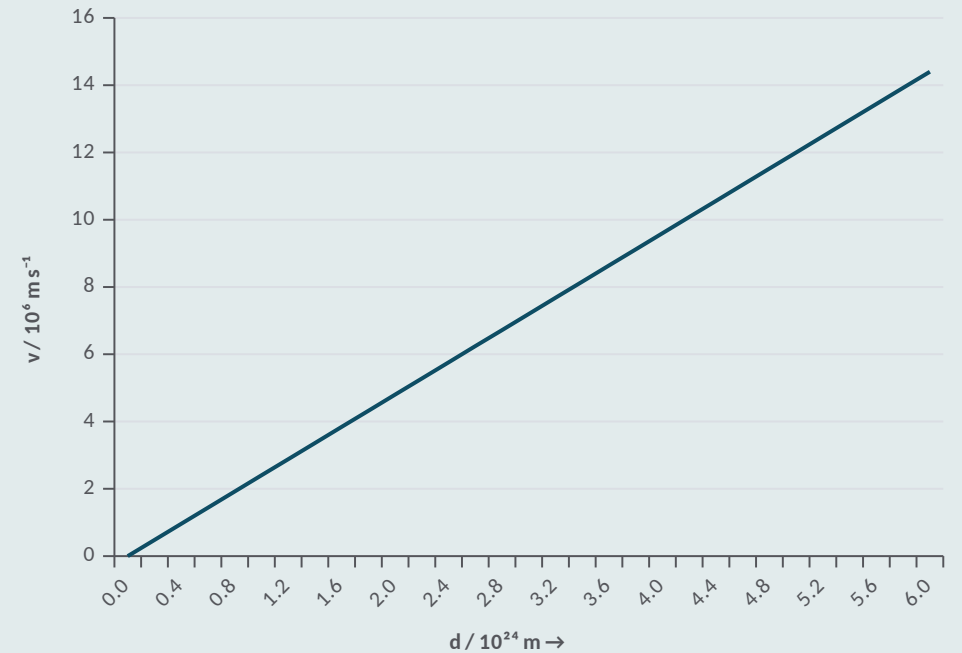
哈勃定律

Hubble found that the recession speed v of a galaxy is **directly proportional to its distance d** — the signature of a uniformly expanding Universe.

星系退行速度与距离成正比——宇宙均匀膨胀的标志。

$$v \approx H_0 d, H_0 \approx 2.4 \times 10^{-18} \text{ s}^{-1} \quad \text{哈勃常数 (教材取值)}$$

- H_0 = gradient of the v - d graph 图线斜率即 H_0
- Unit of H_0 : s^{-1} (often quoted as $\text{km s}^{-1} \text{ Mpc}^{-1}$; 1 Mpc = $3.09 \times 10^{19} \text{ km}$) 单位换算
- All galaxies recede $\Rightarrow$ space itself expands, no centre 膨胀无中心



Recession speed against distance v - d 图

The Big Bang Theory

大爆炸理论

Extrapolating the expansion backwards, the Universe began as an extremely hot, dense point — the **Big Bang** — and has been expanding ever since.

回溯膨胀：宇宙起源于极热极密的起点，此后不断膨胀。

$$t \approx \frac{1}{H_0} = \frac{1}{2.4 \times 10^{-18}} \text{ s} \approx 1.3 \times 10^{10} \text{ yr}$$

宇宙年龄估算

- Evidence: galaxy redshift (Hubble's law) 证据一：红移
- Evidence: **cosmic microwave background** radiation — the cooled relic of the hot early Universe 证据二：微波背景辐射
- Evidence: relative abundance of light elements (H, He) 证据三：轻元素丰度
- $1/H_0$ conversion: $4.2 \times 10^{17} \text{ s} \div (3.16 \times 10^7 \text{ s yr}^{-1}) \approx 1.3 \times 10^{10} \text{ years}$ 秒换算为年

Worked Example 25A

例题解析

WORKED EXAMPLE 25A

The luminosity of the Sun is 3.9×10^{26} W. The Earth orbits the Sun at a mean distance of 1.5×10^8 km. Calculate the radiant flux intensity of the Sun near the Earth.

先把 km 换成 m；此即地球轨道处的 " 太阳常数 "。

SOLUTION 解答

Step 1 Convert distance: $d = 1.5 \times 10^8 \text{ km} = 1.5 \times 10^{11} \text{ m}$

Step 2 Area of sphere: $4\pi d^2 = 4\pi \times (1.5 \times 10^{11})^2 = 2.83 \times 10^{23} \text{ m}^2$

Step 3 $F = L/4\pi d^2 = 3.9 \times 10^{26} / 2.83 \times 10^{23}$

Answer $F \approx 1.4 \times 10^3 \text{ W m}^{-2}$

Worked Examples 25B & 25C

例题解析

WORKED EXAMPLE 25B

A Cepheid variable has luminosity $L = 4.6 \times 10^{15} \text{ W}$ (from its period). Its radiant flux intensity measured on Earth is $F = 1.3 \times 10^{-23} \text{ W m}^{-2}$. Determine its distance from Earth.

1. Rearrange:
2. $d = \sqrt{(4.6 \times 10^{15} / (4\pi \times 1.3 \times 10^{-23}))} = \sqrt{(2.8 \times 10^{37})}$
3. ≈ 560 light-years

Answer $d \approx 5.3 \times 10^{18} \text{ m}$

WORKED EXAMPLE 25C

The wavelength of peak intensity of the Sun's radiation is 510 nm. Use Wien's displacement law to calculate the surface temperature of the Sun.

1. $T = 2.898 \times 10^{-3} / \lambda_{\text{max}}$
2. $T = 2.898 \times 10^{-3} / 510 \times 10^{-9}$

Answer $T \approx 5700 \text{ K}$

Worked Examples 25D & 25E

例题解析

WORKED EXAMPLE 25D

Sirius: luminosity $L = 1.6 \times 10^{28} \text{ W}$, intensity maximum at 290 nm.
Estimate its radius. ($b = 2.898 \times 10^{-3} \text{ m K}$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)

1. Wien: $T = 2.898 \times 10^{-3} / 290 \times 10^{-9} = 1.0 \times 10^4 \text{ K}$
2. $r = \sqrt{(L/4\pi\sigma T^4)} = \sqrt{(1.6 \times 10^{28} / (4\pi \times 5.67 \times 10^{-8} \times 10^{16}))}$
3. $r = \sqrt{(2.25 \times 10^{18})}$

Answer $r \approx 1.5 \times 10^9 \text{ m} (\approx 2 r_{\odot})$

WORKED EXAMPLE 25E

A hydrogen line from Ursa Majoris has wavelength 486.112 nm; in the laboratory it is 486.133 nm. Determine the velocity of the star relative to Earth. ($c = 3.0 \times 10^8 \text{ m s}^{-1}$)

1. $\Delta\lambda = 486.112 - 486.133 = -0.021 \text{ nm}$ (blueshift — approaching)
2. $v = c \Delta\lambda/\lambda = 3.0 \times 10^8 \times (-0.021/486.133)$

Answer $v \approx 1.3 \times 10^4 \text{ m s}^{-1}$ towards Earth

Class Practice

随堂练习

- Q1** The radiant flux intensity of the Sun at Earth is 1400 W m^{-2} . The mean orbital radii of Earth and Mars are $1.5 \times 10^8 \text{ km}$ and $2.3 \times 10^8 \text{ km}$. Determine the flux intensity at Mars.
- Q2** A star has intensity maximum at 650 nm and luminosity $3.0 \times 10^{29} \text{ W}$. Estimate its surface temperature and radius. ($b = 2.898 \times 10^{-3} \text{ m K}$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)
- Q3** The K-line of calcium from galaxy NGC 4889 is measured at 401.8 nm ; in the laboratory it is 393.3 nm . Find the galaxy's speed, direction and distance. ($H_0 = 2.4 \times 10^{-18} \text{ s}^{-1}$)

Answers Q1: $F = 1400 \times (1.5/2.3)^2 \approx 600 \text{ W m}^{-2}$ Q2: $T \approx 4500 \text{ K}$, $r \approx 3.3 \times 10^{10} \text{ m}$ Q3: $v = 6.5 \times 10^6 \text{ m s}^{-1}$ 退行, $d \approx 2.7 \times 10^{24} \text{ m}$

Summary

本章小结

KEY FORMULAS 核心公式

$$F = \frac{L}{4\pi d^2}$$

辐射通量 (反平方)

$$L = 4\pi\sigma r^2 T^4$$

斯忒藩 - 玻尔兹曼

$$v \approx H_0 d$$

哈勃定律

$$\lambda_{max} T = 2.898 \times 10^{-3}$$

维恩位移定律

$$z = \frac{\Delta\lambda}{\lambda} \approx \frac{v}{c}$$

红移

$$t \approx 1 / H_0$$

宇宙年龄

KEY IDEAS 核心思想

- L is intrinsic power; F is what we measure — do not confuse 光度 ≠ 通量
- Standard candle: period $\Rightarrow$ L; L and F $\Rightarrow$ d 标准烛光测距
- Wien + Stefan-Boltzmann together give stellar radius 两定律联立求半径
- Expansion $\Rightarrow$ Big Bang; age $\approx 1/H_0 \approx 13$ billion years 膨胀 $\Rightarrow$ 大爆炸

Exam Tips

备考提示

- Luminosity L (W) vs radiant flux F (W m^{-2}): check units before choosing a formula. L 与 F 单位区分。
- T must be in kelvin in both Wien's and Stefan-Boltzmann laws. 温度一律用 K。
- Unit conversions: $1 \text{ ly} = 9.46 \times 10^{15} \text{ m}$; $1 \text{ Mpc} = 3.09 \times 10^{19} \text{ km}$; convert H_0 consistently. ly 、 Mpc 换算。
- Age of Universe: $1/H_0$ gives seconds — divide by $3.16 \times 10^7 \text{ s yr}^{-1}$ for years. 秒→年换算。